

TIME NUMBERS – 1

Q1

Directions for question 1: Type in your answer in the input box provided below the question.

How many prime numbers are there between 80 and 100?

Your Answer: 3 

Q2



Directions for questions 2 and 3: Select the correct alternative from the given choices.

Which of the following sets of numbers are relative primes?

(a) 51, 85

(b) 26, 65

(c) 57, 76

(d) 29, 75

☐ a) Only (b)

☒ b) Only (d) (Your answer - )

☐ c) Both (a) and (c)

☐ d) All the above

Q3



Directions for questions 2 and 3: Select the correct alternative from the given choices.

If the six-digit number $1457x9$ is divisible by 3, which of the following cannot be equal to x ?

- ☐ a) 1
- ☐ b) 4
- ☐ c) 7

☒ d) 5 (Your answer - ✓)

Q4



Directions for question 4: Type in your answer in the input box provided below the question.

What is the smallest number that must be multiplied with 192 to make it a four-digit perfect square?

Your Answer: 6 ✖

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Solution:

$192 = 3(64)$. The smallest number with which 192 has to be multiplied so that the product is a perfect square is 3. But to get a 4-digit square, we need to multiply by another 4.

$\therefore$ The least number with which 192 has to be multiplied so that the product is a 4-digit square is 12.
Ans : (12)

Directions for questions 5 to 7: Select the correct alternative from the given choices.

How many of the following numbers are divisible by 99?

6006, 9009, 888, 333

☒ a) 1 (Your answer - ✓)

☐ b) 2

☐ c) 3

☐ d) 4

Directions for questions 5 to 7: Select the correct alternative from the given choices.

Which of the following is the largest number which is always a factor of the product of any ten consecutive natural numbers?

☐ a) 1

☒ b) 10^2 (Your answer - ✓)

☐ c) 10^3

☐ d) 10^1

Directions for questions 5 to 7: Select the correct alternative from the given choices.

How many digits are required for numbering the pages of a book containing 1000 pages?

☐ a) 2889

☒ b) 2893 (Correct Answer - ✓)

☐ c) 3000

☐ d) 3126

Solution:

There are 1000 pages in the book.
Page numbers 1 to 9 require 9 numbers i.e. 9 digits.
Page numbers 10 to 99 require $2(90)$, i.e. 180 digits.
Page numbers 100 to 999 require $3(900)$, i.e. 2700 digits.
Page 1000 requires 4 digits.
 $\therefore$ A total of $9 + 180 + 2700 + 4$, i.e. 2893 digits are required.

Choice (B)

Directions for questions 8 and 9: Type in your answer in the input box provided below the question.

Find the units digit of $8858 \times 7234 \times 842 \times 763$.

Your Answer: 2 ✓

Directions for questions 8 and 9: Type in your answer in the input box provided below the question.

When a certain number of tracksuits were distributed equally among a group of four athletes or another group of six athletes, two suits were left in each case. What is the smallest number of suits distributed among each group?

Directions for question 10: Select the correct alternative from the given choices.

What values of n satisfy the following statement (given n is a natural number)?

$2^{3n} - 1$ is divisible by 7.

☐ a) Even values of n

☐ b) Odd values of n

☒ c) All values of n (Correct Answer - ✓)

☐ d) Cannot be determined

Directions for question 11: Type in your answer in the input box provided below the question.

If a number is divided by 15, it leaves a remainder of 7. If thrice the number is divided by 5, what is the remainder?

Your Answer: 1 ✓

Directions for question 12: Select the correct alternative from the given choices.

A is a 3-digit number less than 200. It has no single digit factor except 1. Find the number given that it is not a perfect square.

- ☐ a) 187
- ☐ b) 143
- ☐ c) Either (A) or (B)

☒ d) Cannot be determined (Your answer - ✓)

Directions for question 13: Type in your answer in the input box provided below the question.

The HCF and the LCM of two numbers are 27 and 405 respectively. If one of the numbers is divided by 7, the quotient is 11 and the remainder is four. What is the other number?

Your Answer: 135 ✓

Directions for questions 14 and 15: Select the correct alternative from the given choices.

If p and q are consecutive positive integers (in increasing order), then which of the following is true?

☐ a) $q^2 < p$

☐ b) $2p > q$

☒ c) $(q + 1)^2 > p^2$ (Your answer - ✓)

☐ d) $(p + 2)^3 < q^3$

Directions for questions 14 and 15: Select the correct alternative from the given choices.

What is the greatest negative number that should be added to 1190 to make it a perfect square?

☒ a) -34 (Your answer - ✓)

☐ b) -35

☐ c) -101

☐ d) -1189